



From mice to humans

A multi-omic predictive framework for translational immunology — **detailed report version**, prepared for reading rather than live presentation

Wasim Aluísio Prates-Syed · Lira AA · Cortes N · Silva JDQ · Hamaguchi B · Carvalho E · Castillo-Chávez A · Durães-Carvalho R · Cabral-Marques O · Sabino EC · Krieger JE · Hagan T · Cabral-Miranda G

Laboratory of Genetics and Molecular Cardiology, InCor — FMUSP

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A reading deck, not a talk

- This is the dense companion to the 10-slide lab-meeting deck: same visual system, more text, one topic per slide.
- Class conventions are identical to the reference deck — head blocks, two columns, fig blocks with captions, stats/stat counters, panel comparisons, card blocks and notes.
- Sources: `METHODOLOGY.md` (sections I–XII), `README.md`, `overview.html` and `code-planning.html` on the project site.

Palette #4DBBD5 · #000000 · #4361ee · #A1B0F7

Merriweather headings · Arial body · monospace only for code and accessions

How to read it

- Numbered slides (01 · ...) follow the manuscript logic: question → data → harmonisation → results → modelling → availability → limitations.
- **Bold** marks load-bearing facts; *italics* mark species or emphasis.
- Grey monospace side-notes carry protocols, file names and accessions.
- Every figure keeps the study's own panel caption (Fig. ...) so plots stay traceable to the manuscript.

Modules travel; genes do not.

Blood transcriptomes from matched mouse and human challenge studies were compared across vaccination, acute bacterial infection and sterile injury. At the level of individual orthologous genes the agreement was only moderate, but once the transcriptome was collapsed into Blood Transcription Modules the cross-species signal became strong and stable. Translational accuracy then scaled with the **strength of the perturbation**, and adding **sequence-evolution, transcription-factor and cis-regulatory layers** turned a near-chance mouse baseline into useful prediction.

0.89–0.97

MODULE-LEVEL (BTM
NES) CROSS-SPECIES
PEARSON CORRELATION

0.85

RANDOM-FOREST ROC-
AUC FOR DIRECTIONAL
CONCORDANCE, LOCO

Seven sections, one argument End-to-end pipeline

- **I–II · Question, aims & cohorts** — 6 matched conditions from 201 screened datasets; DMD as a biological negative control.
- **III · Harmonisation** — normalisation, probe collapsing, QC, limma with empirical Bayes, covariate control and downsampling.
- **IV–VI · Results** — granularity, stimulus intensity and kinetics, and the ROC/PR classification benchmark.
- **VII–VIII · Mechanism** — protein-coding evolution and cis-regulatory architecture.
- **IX · Multilayer modelling** — LOCO cross-validation, four algorithms, four translational scores.
- **X–XII · Limits, availability, references.**
- **0 · Data curation** — BioProject / GEO scan and inclusion–exclusion.
- **1 · Standardize** — download, platform normalisation, probe collapse, ortholog mapping.
- **1.1 · Quality control** — arrayQualityMetrics, SA/MA and RLE diagnostics.
- **2 · Differential expression** — limma vs. t-test, combined across species.
- **3 · Comparison** — DGE, GSEA and functional analyses.
- **4 · Performance** — ROC / PR / AUC against DMD and permutation nulls.
- **5 · Evolutionary** — protein sequence and cis-regulatory regulation.
- **6 · Modelling** — tidymodels LOCO mouse-to-human transfer.

Notebook order: 0, 1, 1.1, 2, 3.1, 3.2, 3.3, 4, 5.1, 5.2, 6 · support: required.R, FIT_Exploration_Prediction.Rmd, ImmuneGO_Mouse.Rmd, Tests.Rmd

Mice are the workhorse — but how well do they *actually* predict human immunity?

- Murine models underpin preclinical vaccine research, yet their translational value remains **contested**, contributing to clinical trial failures.
- Prior cross-species comparisons were largely **descriptive**: gene-centric, linear, and disconnected from sequence evolution and cis-regulatory architecture.
- The consequence: **contradictory reports** and no practical framework to prioritise targets for human translation.

201

MOUSE IMMUNIZATION
DATASETS SCREENED
FROM GEO (FIG. 1A)

6

MATCHED MOUSE–
HUMAN CONDITIONS
CARRIED FORWARD

Higher-order structures are conserved even when single genes diverge

One framework, three stimulus categories, matched across species

- **Vaccination** — Fludac® (influenza), Engerix B® (hepatitis B)
- **Infection** — *S. aureus*, *E. coli*
- **Injury** — burn, trauma-haemorrhage
- **+ DMD** (mdx mouse) as a biological negative control

Blood transcriptomes · GEO / BioProject
limma + empirical Bayes · covariates (age, sex, ethnicity)
human downsampled to n = 5, 200 iterations, median statistics

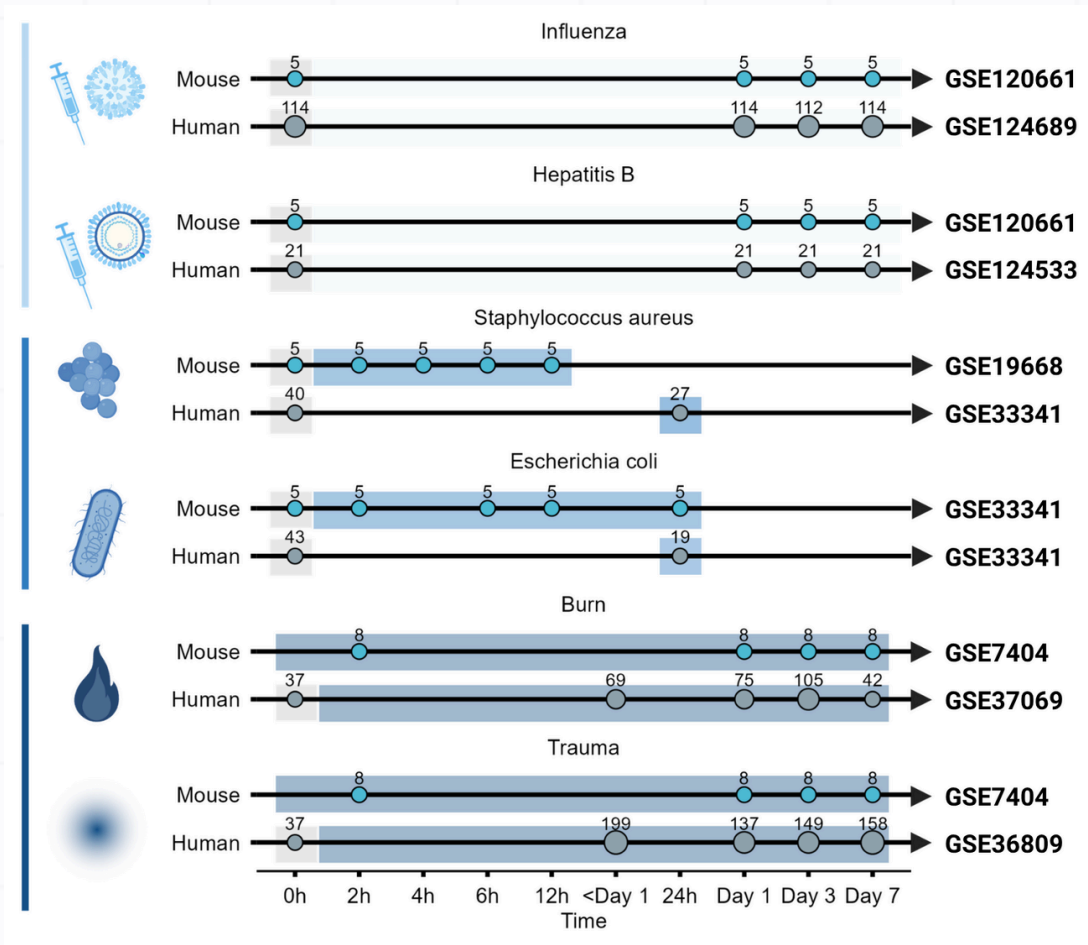


Fig. 1b — experimental timelines for matched mouse and human datasets across vaccination, infection and injury (GSE120661 · GSE124689 · GSE19668 · GSE33341 · GSE7404 · GSE37069 · GSE36809).

From a broad mammalian net to a small matched set

- **BioProject / GEO query** — vaccine terms ("vaccine", "vaccinated", "vaccination", "vaccines") combined with "transcriptome gene expression", "material transcriptome", "capture_whole" and "org mammals", **excluding** *Homo sapiens*.
- **Organism restriction** — a second query isolated *Mus musculus*, excluding tumour, cancer and autoimmune records.
- **Assay restriction** — only "Expression profiling by array" and "Expression profiling by high throughput sequencing".

Manual annotation is the rate-limiting step

- Only studies of **vaccines against human pathogens** were kept; BioProject and GEO records were merged via **PRJNA** accessions and split by **vaccine, sample source and RNA protocol**.
- Datasets with partial methods, or BCR/TCR repertoire only, were excluded.
- Records were annotated with GEOquery and curated against the publication, capturing source, condition, antigen, vaccine type, adjuvant, time points, doses, route and assay.

Where records conflicted with the publication, the GEO record was prioritised

Only a handful of murine cohorts met the bar for a direct comparison

201

MOUSE IMMUNIZATION
DATASETS RETRIEVED
FROM GEO

6

MATCHED MOUSE–HUMAN
CONDITIONS CARRIED
FORWARD

GSE120661 (influenza + hepatitis B) was carried forward; GSE182858 (influenza in “dirty” mice) could not be compared with the human cohort — sampling at 3 h before/after immunisation vs. days 1, 3 and 7 — and was not used

Vaccination & infection

Vaccination — subunit / inactivated

- **Influenza (Fluad®, TIV + MF59)** — human GSE124689 · mouse GSE120661
- **Hepatitis B (Engerix B®)** — human GSE124533 · mouse GSE120661

Acute bacterial infection

- ***S. aureus* bacteraemia** — human GSE33341 · mouse GSE19668
- ***E. coli* sepsis** — human GSE33341 · mouse GSE33341

Injury & negative control

Sterile injury

- **Severe burn** — human GSE37069 · mouse GSE7404
- **Trauma-haemorrhage** — human GSE36809 · mouse GSE7404

Biological negative control

- **Duchenne muscular dystrophy (mdx)** — mouse GSE1025, muscle tissue, peak pathological timepoint day 28.

GSE120661 — one series, four arms, six time points

- **Time points:** 4, 8, 24, 48, 72 and 168 h (4 h → day 7).
- **Samples per arm:** 5; **single dose**; intramuscular route.
- **Tissue / assay:** PBMCs, microarray.

Arm	Pathogen	Vaccine	Adjuvant
1	HBV	Engerix-B	Alum
2	HBV	Engerix-B	None
3	Influenza	Agrippal	None
4	Influenza	Fluad®	MF59

Why this cohort matters

- The four arms separate **pathogen, antigen, adjuvant and vaccine type** within a single experiment — a clean within-series contrast.
- Human counterparts are **prime-boost** regimens, whereas the mouse arms are **single-dose**, so the adaptive trajectory is not strictly comparable.
- The adjuvant contrast (MF59 vs. none; alum vs. none) is what lets the study separate *intensity* from *identity*.

GSE182858 (Quadri 2019-2020, IN route, ±Addavax) and GSE224584 (N. meningitidis ACWY, whole blood) illustrate the screened-but-unused periphery

A perturbation that should *not* transfer to human blood

- **Model:** mdx mouse, Duchenne muscular dystrophy; peak pathological timepoint **day 28**; muscle tissue (GSE1025, Affymetrix Murine U74A v2).
- Used to benchmark **specificity**: a classifier built on a disease that is neither an immune challenge nor a blood phenotype should perform near chance.
- The DMD model anchors the “negative” side of every ROC curve, so that high AUCs for infection and injury can be attributed to conservation rather than to methodological artefacts.

DMD ROC-AUC 0.48–0.61 — near random, as expected

Two more control ideas on the record

- The performance framework also compares against a “**perfect**” **model** that exactly replicates the human truth labels — the upper bound of the AUC.
- **Permutation null distributions** accompany the biological controls in the performance notebook (`4_Performance_DifferentTimepoints.rmd`).
- Together these three references (truth, DMD, permutation) let each reported AUC be read as a position on a continuum from chance to perfect transfer.

One normalisation scheme per technology

- **Affymetrix oligonucleotide arrays** (e.g. Human Gene 1.0 ST): raw probe intensity .CEL files were preprocessed with **RMA** via `affy / oligo` — background correction, quantile normalisation and median-polish summarisation.
- **Illumina BeadChips & Agilent arrays**: between-array quantile normalisation (`limma::normalizeBetweenArrays(method = "quantile")`) to align empirical distributions while preserving biological rank order.
- Expression matrices were **log2-transformed** before modelling.

Why not a single method

- RMA is designed for the multi-probe structure of Affymetrix arrays; forcing it onto bead-level Illumina data, or quantile-normalising .CEL files before RMA, would violate each platform's noise model.
- The goal is **comparability of distributions**, not identity of algorithm: each cohort is normalised in its own native space, then mapped to orthologs.
- All preprocessing lives in `1_Download_Standardize_Datasets.Rmd`.

log2 intensity retained; no arbitrary truncation of effect sizes before modelling

One probe per gene, one ortholog per human gene

- **Probe collapsing:** probes mapping to the same **Entrez ID** were collapsed by selecting the probe with the **highest median expression** — not averaging, which would dampen signal across non-responsive probes.
- **Human reference selection:** when several genes mapped to the same ortholog, the Entrez ID with the **highest median expression** was used as the reference.
- **Orthology:** protein-coding genes with **one-to-one orthology to human** from **MGI**; genes with multiple orthologs were **excluded**.

Annotation sources & baseline

- Platform annotation via Bioconductor packages (`illuminaHumanv4.db`, `hugene10sttranscriptcluster.db`) and `biomaRt`.
- Collapsed genes and their log2-fold changes enter the model as a **baseline contrast against pre-treatment** (0 h / day 0).
- The one-to-one requirement trades coverage for interpretability — it is also a limitation, because it discards duplicated and many-to-many orthologs.

Probes: highest median, not mean · Orthologs: MGI 1:1 only ·
Human ref: highest median Entrez

Three complementary views of array quality

- **Distribution diagnostics:** density plots, boxplots and scatterplots to inspect normalisation and between-sample comparability (`arrayQualityMetrics`, `limma`).
- **PCA:** used to spot outliers and batch effects. A subset of **human trauma samples** separated significantly and was **excluded** — the metadata could not identify a covariate cause, so it was hypothesised to reflect unreported trauma severity.
- **RLE:** $RLE_{gi} = \log_2(E_{gi}) - \text{median}_j(\log_2(E_{gj}))$; samples with anomalous IQR divergence or median shifts were flagged and removed where technical confounding was confirmed.

The RLE statistic

$$RLE_{gi} = \log_2(E_{gi}) - \text{median}_j(\log_2(E_{gj}))$$

- A well-normalised sample has RLE medians near zero and comparable IQRs.
- The trauma exclusion is disclosed rather than silently dropped: it is a reminder that clinical metadata can be incomplete.
- QC outputs and diagnostics were archived (`tables/QualityControl/`) so the decisions are reproducible.

1.1_QualityControl.Rmd · SA and MA plots · before-vs-processed summaries

Empirical Bayes moderation, condition by condition

- **Low-abundance filter:** retain genes with **log2-intensity > 6** in at least five samples.
- **Model:** an unintercepted design matrix, $Y_g = X \beta_g + \varepsilon_g$, with X parameterised by organism, stimulus and timepoint.
- **Contrasts:** each post-treatment timepoint against its matched pre-treatment baseline (e.g. day 1 – day 0, day 3 – day 0, day 7 – day 0).
- **Variance shrinkage:** `limma::eBayes(fit, trend = TRUE, robust = TRUE)`, squeezing gene-wise variances toward an intensity-dependent prior.

Why moderation helps here

$$\tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g}$$

- trend accounts for the mean–variance relationship; robust limits the pull of outlier genes.
- Moderated t -statistics feed everything downstream: effect sizes, ranks, one-sided p-values and pathway enrichment.
- Notebook: `2_Differential_Gene_Expression.Rmd`.

DGE threshold: BH-adjusted $P < 0.05$ · effect sizes retained continuously

Adjusting the clinical noise floor What this buys, and what it costs

- **Longitudinal designs:** individuals treated as **random effects**.
- **Fixed effects:** **sex, ethnicity and age** were included in the limma design matrix for both longitudinal and case-control human datasets.
- **Batch:** for the *E. coli* mouse dataset, weight estimation was stratified by experimental batch (`var.group = batch`) to keep batch effects out of the biological signal.
- **Heteroscedasticity:** sample-quality weights via `limma::arrayWeights`, so high-variance samples contribute less to the fit.
- Covariate control is the main reason the study can compare clinical human cohorts — often unbalanced by design — with inbred, age-matched mouse groups.
- It cannot fix **unmeasured** confounding: burn and trauma severity scores are absent, so injury comparisons carry irreducible uncertainty.
- `arrayWeights` protects the model from a few noisy arrays without discarding them outright.

E. coli: mouse and human blocks come from the same GEO series (GSE33341) → batch handled explicitly

The choice changes the conclusion

- The pipeline was compared against conventional **paired t-tests** used in prior studies (Supplementary Fig. S2).
- With **empirical Bayes variance shrinkage** plus adjustment for demographic covariates and batch effects, **limma clearly outperformed** the standard paired t-test.
- Decisively, the t-test approach **failed to identify significant DEGs in the vaccinated mouse cohorts** — which would have produced the erroneous conclusion that cross-species conservation is limited.

The methodological point

- Weak stimuli leave small effects; a method that cannot detect them will under-report conservation and bias the whole narrative toward “mice don’t translate”.
- Moderated statistics recover these weak but coordinated signatures, which is what makes the vaccination result interpretable rather than merely null.
- This is a reproducibility lesson as much as a statistical one: the tool choice, not just the data, shapes the headline.

limma: Ritchie et al., Nucleic Acids Research 2015 · eBayes trend + robust

Equalising an unbalanced comparison

- Human cohorts were much larger than the five-sample murine arms, so human samples were **downsampled to $n = 5$** .
- A **bootstrap over 200 iterations** was run, and the **median** of the statistics derived from limma was taken.
- This removes pseudo-significance: a large human n would otherwise give tiny p-values for trivial effects and inflate apparent cross-species agreement.

Why median, not mean

- The median is robust to the occasional bootstrap draw with an extreme effect, which would skew a mean across iterations.
- Sampling stability itself becomes a diagnostic — it is reported alongside the divergence metrics in `3.1_DGE_analyses.Rmd`.
- The same balancing logic underpins the inverse-variance weighting used later ($1 / (SE_H^2 + SE_M^2)$).

$n = 5 \cdot 200$ bootstrap iterations · median statistics · human downsampling

Two thresholds for two purposes

- **Differential expression:** genes were called DEGs when the **Benjamini– Hochberg adjusted $P < 0.05$** . Effect-size magnitude ($\log_2FC$) was kept continuous, with no arbitrary truncation, to power rank-based pathway tests.
- **Enrichment:** the GSEA FDR limit was set at **$padj \leq 0.25$** , deliberately looser to capture coordinated pathway trends — especially for weaker stimuli such as vaccination.
- Raw p-value distributions were inspected for every contrast.

What the diagnostics revealed

- Most conditions showed the expected enrichment of small p-values ($p < 0.05$).
- Where adjusted p-values yielded few genes — some murine injury models — raw histograms **lacked the expected peak at zero** or trended conservatively, indicating lower power or high intra-group variability.
- Murine **burn and trauma** datasets showed near-**uniform adjusted p-value distributions**, reflecting the high inter-individual variance of cross-sectional injury studies versus the longitudinal vaccine and infection designs.

Thresholds are analysis-level, not evidence-level: $padj \leq 0.25$ is exploratory for modules

Collapsing the transcriptome into immune modules

- Primary unit: **346 Blood Transcription Modules** (Li et al., *Nat Immunol* 2014), capturing leukocyte subsets (T, B, NK, monocyte, neutrophil, dendritic) and functional states (interferon, inflammatory chemokines, cell cycle).
- Modules are grouped hierarchically into **16 physiological domains**.
- For each contrast, orthologous genes were **pre-ranked** by $\text{Rank}(g) = \log_2\text{FC}_g \times (-\log_{10}(\text{adjusted } P_g))$.
- GSEA via **clusterProfiler**; scores normalised for gene-set size (**NES**). MSigDB Hallmarks ran in parallel as an independent reference.

From NES to cross-species concordance

- **Module-level** concordance = **Pearson correlation on NES**, weighted by $-\log_{10}(\text{adjusted } P)$ from GSEA.
- **Gene-level** concordance = **weighted Spearman ρ** , weighted by the inverse standard error ($1/\text{SE}$) of murine effect sizes.
- Correlations were also computed on nested module subsets — all enriched modules, mouse-significant modules, and dual-significant modules — to test how much stringency changes translatability.

3.2_Compute_GSEA.Rmd · 3.3_Functional_Analyses.Rmd · padj ≤ 0.25

0.89–0.97

- Gene-level effect sizes correlate only **moderately**; module activation is **highly conserved**.
- Shared leading-edge genes = **67.5–79%** of mouse LEGs in enriched modules (**70.5% pooled**).
- Collapsing dimensionality into BTMs **reduces noise** and exposes a robust translational signature.

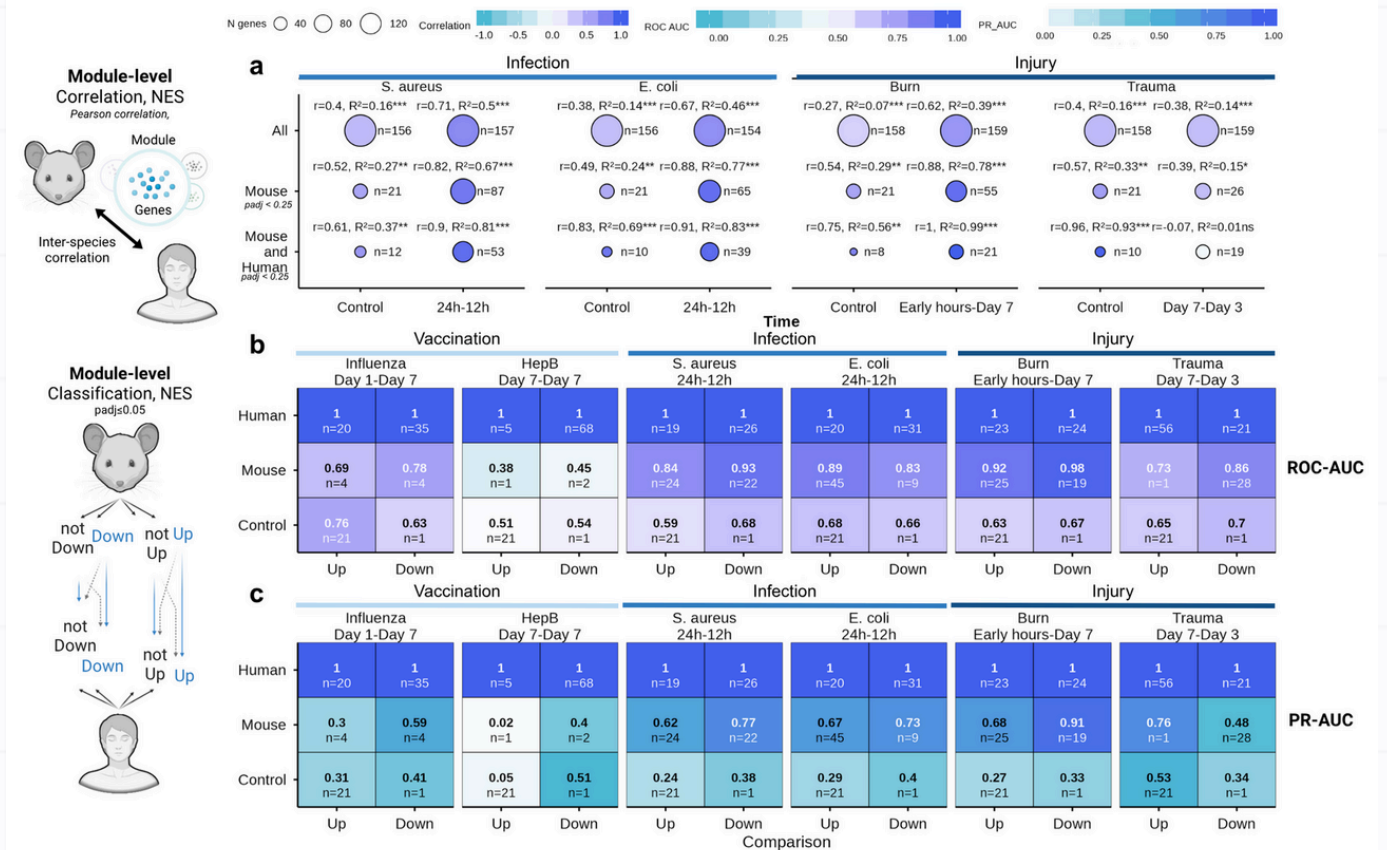


Fig. 4a–c — BTM module-level correlation (a) and classification performance (b, c) across conditions. · Fig. 5a — shared leading-edge genes across BTM process categories.

The core of a module is what transfers

- For every enriched BTM ($\text{padj} \leq 0.25$), **Leading-Edge Genes (LEGs)** — those accounting for the core enrichment signal before the peak running enrichment score — were extracted for both species.
- Module genes were partitioned into **Shared LEGs, Human-specific LEGs, Mouse-specific LEGs** and **Non-LEGs**.
- **Shared LEGs = 67.5–79%** of mouse LEGs in significant enriched modules (**70.5% pooled**).
- Shared LEGs showed higher expression and **lower 1/SE** than not-shared LEGs (WMW adjusted $P < 2.22 \times 10^{-16}$ and $P < 0.00058$), though 1/SE did not differ in burn and trauma ($P = 0.62$ and 0.39).

Rank conservation in a core module

- Intra-module prioritisation was tested on the “*immune activation — generic cluster*” (**BTM M37.0**; 347 genes): innate inflammatory, Toll-like receptor and chemokine programs.
- For each gene, $\text{Rank Score}_g = \log_2(\text{FC}_g) \times (-\log_{10}(P_g))$; genes were ranked within species and compared by **Spearman correlation**.
- A parallel-axis plot connects human and mouse positions, colouring Shared / species-specific / Non-LEGs.

Fig. 5a — LEG proportions · Fig. 5b — rank-conservation trajectory

Translatability scales with the *strength* of the perturbation

Infection & injury — conserved

$\rho = 0.41\text{--}0.59$

Acute *S. aureus*, *E. coli* and burn engage evolutionarily stable, highly conserved programs.

Effect-size difference ($|\Delta\log_2FC|$) significantly smaller than negative controls —
WMW adjusted $P < 1 \times 10^{-15}$.

Single-dose vaccination — diverges

$\rho = -0.32\text{--}0.27$

Fluad® and Engerix B® fail to reach the systemic activation threshold, remaining near baseline controls.

Preclinical immunogenicity may require **boosters or stronger adjuvants** to emerge from species-specific noise.

Concordance measured as weighted Spearman ρ on cross-species effect sizes; strength of stimulus, not species identity, sets the ceiling

Chronology is the wrong clock

- Mouse samples are collected **2–6 h** after challenge; the human peak inflammatory window is **days 1–3**.
- These are biologically equivalent phases of the response even though the elapsed time differs by an order of magnitude.
- Conclusion: align timepoints by **biological equivalence**, not by calendar time.
- Human injury times (recorded in hours) were converted to days before binning.

How timepoints were binned

- Samples were discretised to match the mouse collection schedule: **< 1 day** → Early Hours; **1–2 d** → Day 1; **2–5 d** → Day 3; **5–10 d** → Day 7; **10–18 d** → Day 14; **18–25 d** → Day 21; **> 25 d** → Late.
- Subjects were grouped into **Trauma**, **Burn** and **Control** using subject identifiers.

Mouse 2–6 h $\approx$ human days 1–3 · cross-temporal ROC tests explicitly sample different timepoints

Turning DGE into a prediction problem

- Limma produces **two-sided** p-values; **one-sided** p-values were recalculated from the moderated *t*-statistics — upper tail for upregulated genes, lower tail for downregulated genes — then BH-adjusted.
- For each gene and timepoint, the **prediction probability** was 1 – adjusted one-sided P; **ROC** and **precision–recall** curves were built and the **AUC** computed (pROC, yardstick).
- The **human dataset is the reference “truth”**; mouse predictions are scored against it.

Controls that give the AUC meaning

- A “**perfect**” **model** replicates the human labels exactly — the ceiling.
- A “**negative control**” **model** is built from the **DMD** (*mdx*) expression table — the floor.
- **Permutation null distributions** provide a further reference.
- The same framework was applied at **module level**, using NES and $-\log_{10}(\text{adjusted } P)$ from GSEA.

4_Performance_DifferentTimepoints.rmd · genes and BTMs ·
ROC + PR

Mouse signatures call human direction — best at *module* level

0.84–0.98

ROC-AUC · MOUSE
BTM SIGNATURES
CLASSIFYING HUMAN
MODULE REGULATION

0.64–0.74

ROC-AUC · GENE-
LEVEL MOUSE
SIGNATURES IN
INFECTION & INJURY

Biological negative control (DMD): ROC-AUC 0.48–0.61
— near random

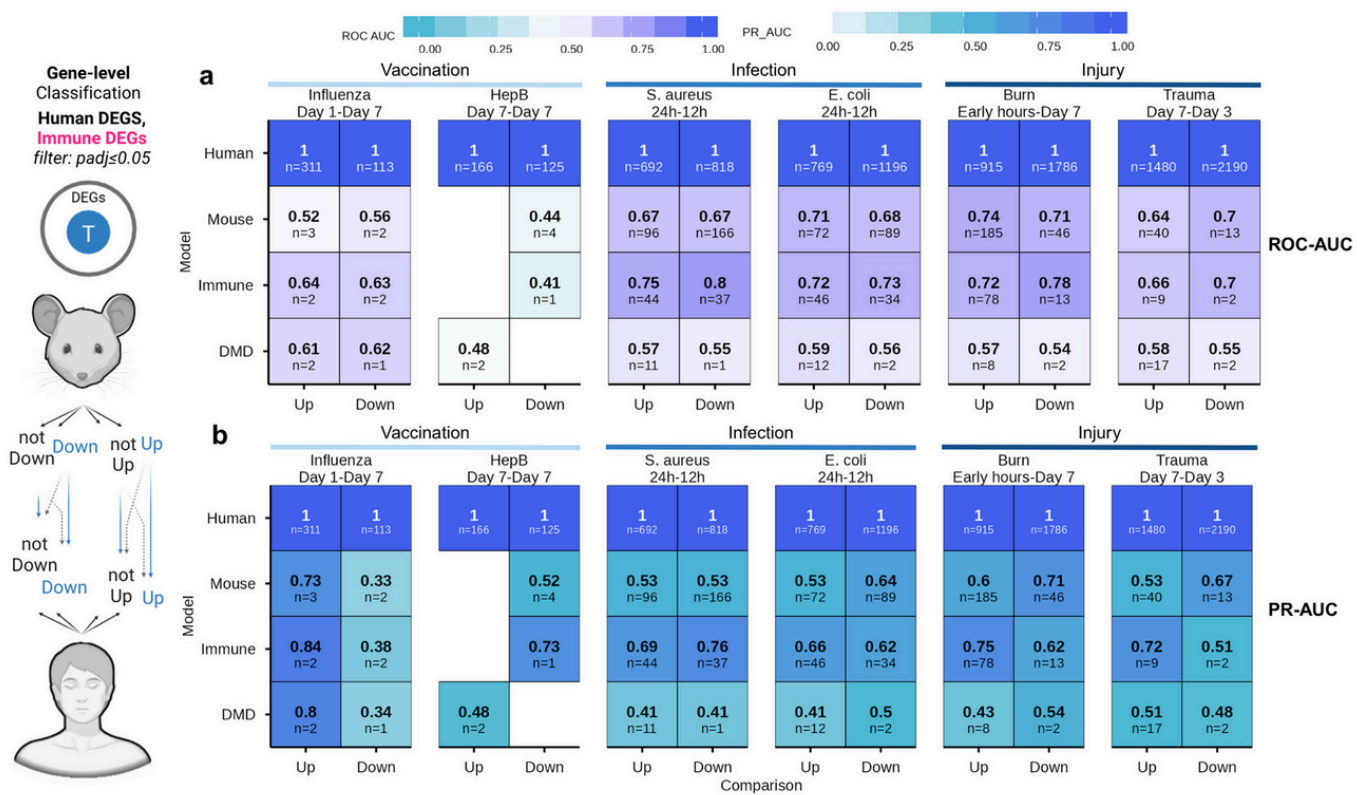


Fig. 3a–b — ROC-AUC and PR-AUC matrices for mouse signatures classifying up- and downregulated human genes, with human self-prediction and the DMD negative control.

Coding constraint buys only a small amount of stability

- **Greater protein sequence identity** is associated with **lower** $|\Delta\log2FC|$ — Spearman $\rho = -0.08$.
- **Greater Kimura (K80) distance** is associated with **higher** $|\Delta\log2FC|$ — $\rho = 0.10$.
- The per-gene effects are small but consistently signed: coding constraint tracks expression stability.
- Sequences from Ensembl via biomaRt (confidence score = 1); **longest coding sequence** kept per gene; pairwise **global Needleman–Wunsch** alignment (pwalign); **K80** distance (ape::dist.dna), with **JC69** as a control.
- Protein identity came from Ensembl BioMart homology tables.

K80 $d = -\frac{1}{2} \ln(1 - 2P - Q) - \frac{1}{4} \ln(1 - 2Q)$ · longest-isoform canonical CDS

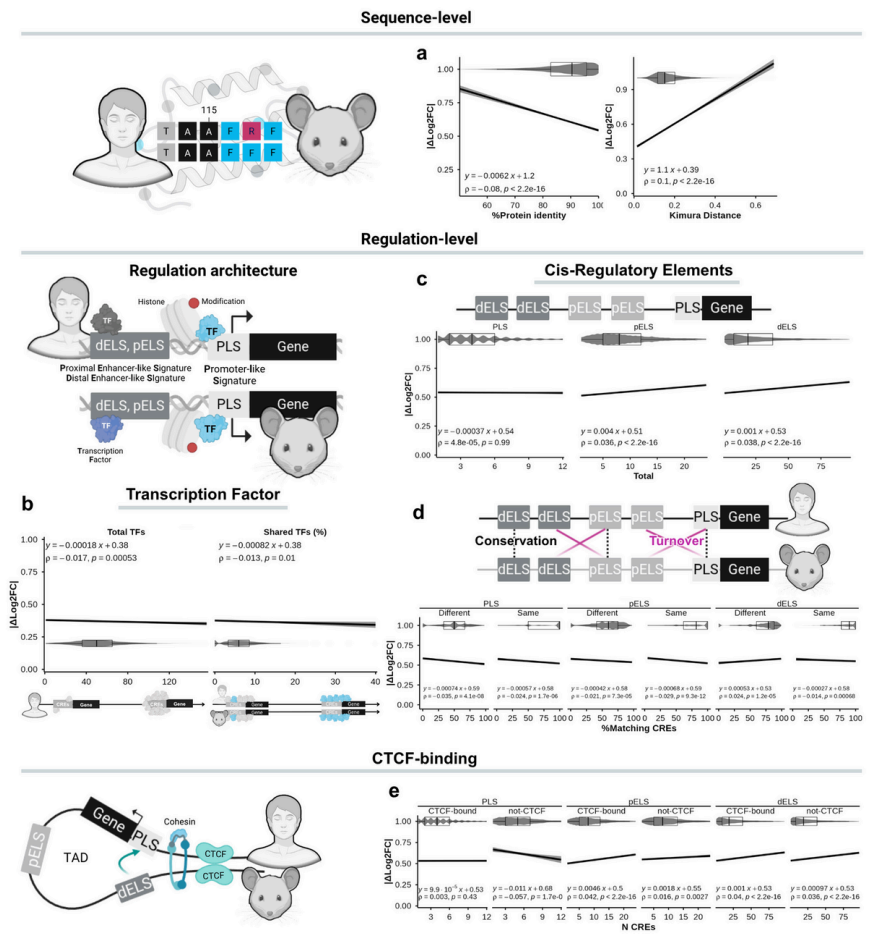


Fig. 6a–e — protein sequence identity and Kimura distance (a), transcription factor availability (b), cis-regulatory element abundance (c), CRE conservation and turnover (d) and CTCF binding (e) against $|\Delta\log2FC|$.

Regulatory plasticity, not coding change

ENCODE cCREs, GRCh38 vs. mm10

- **TF availability and TF sharing** correlate **negatively** with $|\Delta\log_2FC|$ — $\rho = -0.017 / -0.013$.
- **Enhancer multiplicity** (pELS / dELS counts) correlates **positively** — $\rho = 0.036 / 0.039$.
- **CRE turnover and CTCF-bound pELS** correlate positively — $\rho = 0.042$.
- **Promoter-like signatures (PLS)** show **no significant relationship**.
- Fitted with OLS and tested with monotonic Spearman rank correlations.
- Human and mouse **candidate cis-regulatory elements** were taken from ENCODE via the SCREEN portal, together with homologous cCREs.
- Human cCREs were classed as **PLS**, **pELS** or **dELS**, then split by **CTCF-bound vs. CTCF-non-bound**.
- cCREs were assigned to their associated genes, then compared to annotated murine elements; matches were counted per class and CTCF status.

tables/Genomic cCRE BEDs ·

5.2_EvolutionaryAnalysis_Regulation.Rmd

Associated with lower $|\Delta\log_2FC|$ (convergence)

- Higher protein sequence identity $\rho = -0.08$
- Greater TF availability and TF sharing $\rho = -0.017 / -0.013$
- Conserved promoter-like (PLS) signatures $n.s.$

Associated with higher $|\Delta\log_2FC|$ (divergence)

- Greater Kimura (K80) evolutionary distance $\rho = 0.10$
- Enhancer multiplicity — pELS / dELS counts $\rho = 0.036 / 0.039$
- CRE turnover and CTCF-bound pELS $\rho = 0.042$

Sequence constraint tracks *stability*; regulatory plasticity tracks *divergence*

Two nested layers; predict the human response gene by gene

- **DGE Baseline** — the mouse differential-expression summary only: mouse log2FC, 1/SE, and mouse absolute rank (overall and within BTMs).
- **Full + BTM** — adds structural coding evolution (identity_human2mouse, dist_k80), cis-regulatory features (PLS/pELS/dELS counts and match percentages, CTCF-bound fractions), TF features (n_tf_total, pct_tf_shared) and BTM membership (immune vs. non-immune).
- Two **framings**: **A (predictive)** keeps the mouse magnitude; **B (explanatory)** removes it to test the biological layers alone.

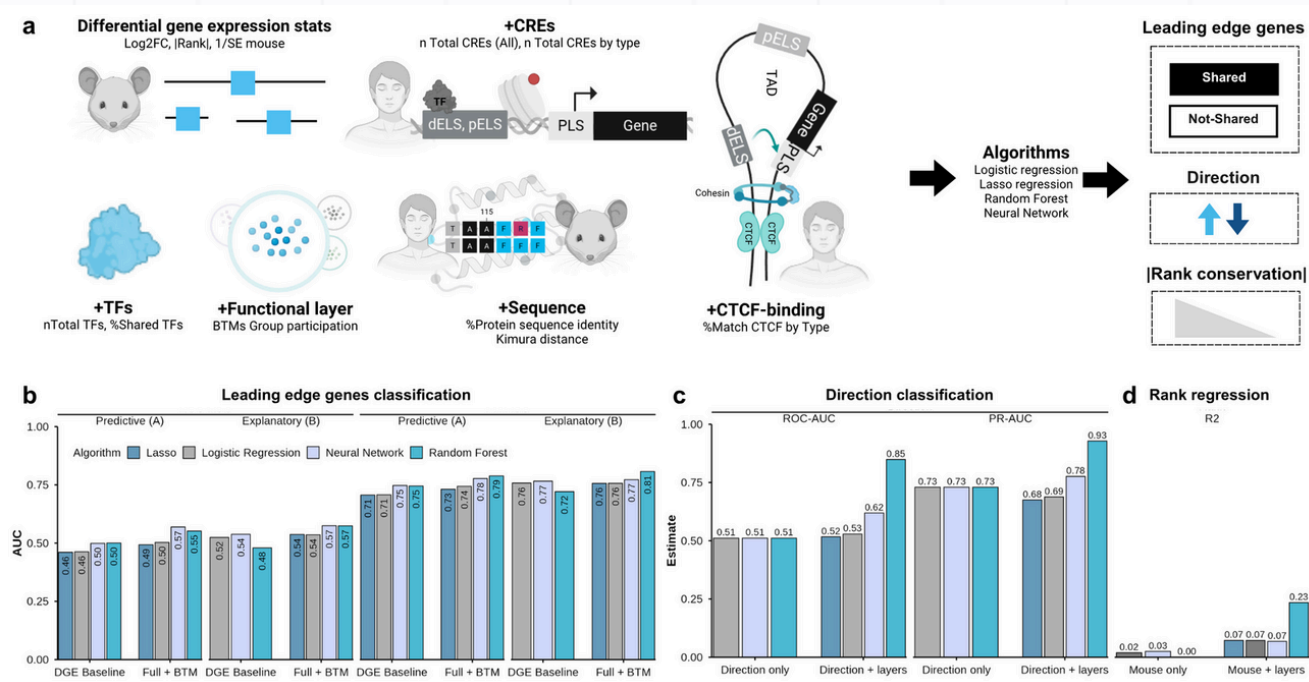


Fig. 7 — multilayer modelling: (a) layer integration workflow, (b) shared-LEG classification, (c) directional concordance and (d) human rank regression, across algorithms and feature layers.

Leave-one-condition-out, so no gene is scored by its own pathogen

- Models are evaluated by **leave-one-condition-out (LOCO)** over the four infection/injury folds — *S. aureus*, *E. coli*, burn, trauma.
- **Every gene is predicted by a model trained without its condition**, producing the out-of-fold predictions that feed both the metrics and the exported scores.
- Two **universes**: all human DEGs ($p_{\text{adj_human}} \leq 0.05$) for rank transfer and direction; mouse LEGs (Shared vs Mouse only) for shared-LEG classification.
- **Targets**: rank_human (0–100), sign_concordant, and Shared vs Mouse only.

Four algorithms, identical recipes

- **Linear / logistic regression** — the baseline.
- **Lasso** — glmnet, mixture = 1, penalty 0.01 (skipped when fewer than two predictors remain).
- **Random forest** — ranger, 500 trees, mtry = 8, min_n = 5.
- **Neural network** — nnet, one hidden layer of 10 units, 200 epochs, weight decay 0.01.
- Shared recipe steps: step_dummy, step_zv/step_nzv, step_corr (classification, threshold 0.9) and step_normalize.

Feature importance: log-odds (logistic) · standardised effects (linear) · lasso coefficients · Gini (RF)

Layers turn near-chance transfer into *useful* prediction

- Mouse rank alone: $R^2 = 0.001$ · mouse direction alone: **ROC-AUC = 0.511** — no transfer.
- Adding evolution, TF, CRE and BTM layers → random forest $R^2 = 0.2345$ for human rank and **ROC-AUC = 0.8488** for direction.
- Shared vs. mouse-only LEGs stay hard: lasso best by a whisker — **0.6173** predictive / **0.6271** explanatory.

LOCO-CV (4 folds) · nested layers: DGE Baseline → Full + BTM
4 algorithms: linear/logistic · lasso · random forest · neural network

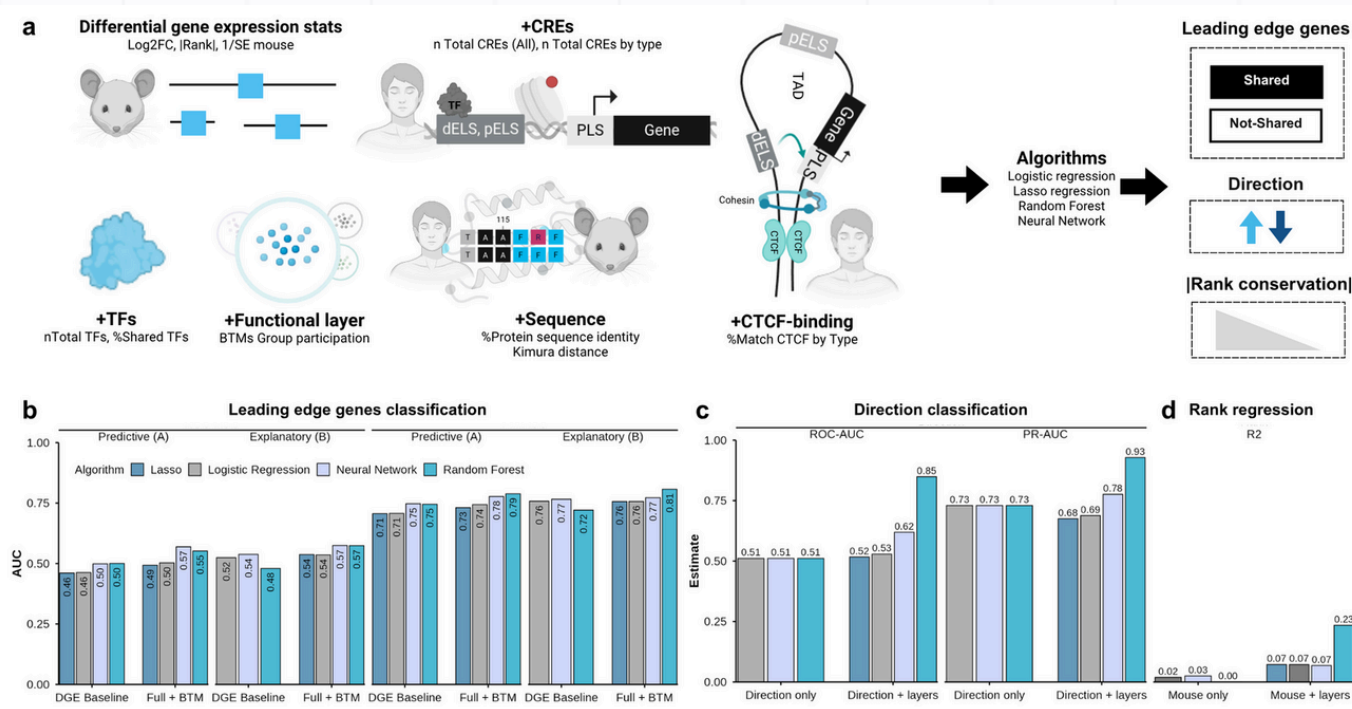


Fig. 7b–d — shared-LEG classification (b), directional concordance (c) and human-rank regression (d) across algorithms and feature layers under LOCO cross-validation.

The algorithm ranking is itself a result

Consequences for practice

- For **rank transfer** and **direction**, the random forest is best by a wide margin (R^2 0.2345 vs. 0.072 lasso/linear; ROC-AUC 0.8488 vs. 0.517–0.619).
- For **shared-LEG classification**, the lasso is marginally best (0.6173 vs. 0.601 linear predictive; 0.6271 vs. 0.611 explanatory) and the gap is tiny.
- Interpretation: the transferable signal for **rank and direction is predominantly non-linear and multivariate**, whereas **LEG sharing is captured by a linear, modular** signal.
- Mouse-only features reduced every algorithm to **near chance** for direction — the biology, not the algorithm, is doing the work.
- If you want a *rank* or a *direction* prediction, use a tree ensemble with the full layer set.
- If you want to know whether a gene is a *shared leading-edge* member, a simple penalised linear model is enough.
- Adding the BTM context is what lifted the random forest to $R^2 = 0.23$ and ROC-AUC = 0.85 — module membership is a feature, not just a summary.

The lasso improved only marginally over unregularised linear models — regularisation is not the source of the gain

Four complementary out-of-fold scores

Two axes, two uses

- `score_shared` — probability of being a **shared LEG**.
- `score_rank` — **predicted human absolute rank** (0–100).
- `score_direction` — probability of **concordant direction**.
- `score_translational` — `score_rank` × `score_direction`.

Every gene is scored by a model trained without its condition (out-of-fold LOCO)

- An **observed prioritisation score** multiplies cross-species convergence by the **observed** directional concordance, ranking genes measured in both species.

- The scores capture **observed cross-species agreement** versus **predicted human response magnitude** — two different questions that a single ranking would conflate.
- `score_translational` is the forward-looking one: given new murine data, it forecasts the human response.
- Derived metrics support interpretation: `rank_diff` = `|rank_mouse - rank_human|` (**convergence**) and `dual_rank` = `min(rank_mouse, rank_human)` (**relevance**).
- A ranked candidate list is exported as `priority_relevant_convergent_top20.csv`.

Outputs in `Modelling/Tables/` ·
`observed_vs_predicted.csv/.rds`

Trained models you can reload

- Best workflows are reduced with **butcher()** and stored with `compress = "xz"` — `rf_model_{shared,rank,direction}.rds`, `nn_model_{shared,rank,direction}.rds`, `lasso_model_shared.rds`.
- Apply to a new mouse experiment with a single call:

```
1 model_rank <- readRDS("Modelling/Models/rf_model_rank.rds")
2 predict(model_rank, new_data = my_features)
```

Archived artefacts

- **Metrics:** `metrics_shared.rds`, `metrics_rank.rds`, `metrics_direction.rds`, `metrics_shared_optionB.rds`.
- **Coefficients & importance:** `coefs_*.rds`, `imp_*.rds`.
- **Out-of-fold predictions:** `pred_shared.rds`, `pred_rank.rds`, `pred_direction.rds`.
- **Tables:** `shared_loco.csv`, `shared_loco_optionB.csv`, `rank_loco.csv`, `direction_loco.csv`, `observed_vs_predicted.csv`, `priority_relevant_convergent_top20.csv`.
- **Figures:** `Fig7_multilayer_modelling.png` and performance bars.

Modelling/{Models,Tables,Figures} · written by
6_Statistical_Modelling.Rmd

Open code, trained models, and a notebook to *apply* them

PIPELINE

mousetohuman_multilayer

Full R Markdown pipeline — curation, QC, limma DGE, GSEA, evolutionary and regulatory analyses, and the LOCO modelling script. Pinned with renv.

github.com/wapsyed/mousetohuman_multilayer

DOCUMENTATION

Project site

Overview, complete methodology — including the multilayer modelling pipeline — and the notebook map, in one place.

wapsyed.github.io/mousetohuman_multilayer

APPLY IT

mousetohuman_predict

Step-by-step notebook: from DGE input to GSEA, model prediction and visualisation of the four translational scores. Runs in RStudio via Binder.

github.com/wapsyed/mousetohuman_predict

R ≥ 4.5.2 · Bioconductor 3.22 · MIT license · `renv::restore()` reproduces the environment

Cohort & experimental limits

- **Small murine cohorts** constrained statistical power; scarce harmonised metadata limited generalisability.
- **Two inbred strains only** — C57BL/6 and CB6F1; because host genetics shapes translatability, low correlation in some modules may reflect **strain–hypothesis mismatch** (Th1-biased C57BL/6 vs. Th2-biased BALB/c) rather than species-level failure.
- **Single-dose mouse vaccination** vs. human **prime-boost** regimens limits comparison of adaptive trajectories.
- Lab mice are **immunologically naïve**, whereas humans carry complex exposure histories (evident in the Fluad® data).
- Human burn/trauma datasets lacked standardised **severity scores**.

Scope, resolution & caveats

- Analyses rely on Ensembl-annotated **one-to-one orthologs**; the ~50% amino-acid identity baseline does not capture domain topology, PPI rewiring or the functional impact of mutations.
- Scores derive from **blood-based BTMs** and need testing in other tissues and gene sets (MSigDB, Gene Ontology).
- **Not yet included:** regional constraint (PhastCons/phyloP), transposable elements, associated TF expression; baseline expression, gene length, GC content, CRE alignment and substitution-rate acceleration were not controlled.
- **Accession caveat:** the *E. coli* mouse and human blocks both come from **GSE33341** (one series with mouse and human blocks), so it is not an independent cross-species accession — batch was handled explicitly via `var.group`.

Translatability is a function of *granularity*, *intensity* and *timing*

What the study establishes

- Module-level analysis beats gene-level analysis for cross-species concordance (BTM NES $\rho = 0.89$ – 0.97).
- Acute infection and injury translate; single-dose vaccination does not ($\rho = 0.41$ – 0.59 vs. -0.32 – 0.27).
- Evolutionary and regulatory layers carry a conserved signal beyond mouse DGE — lifting the random forest to $R^2 = 0.2345$ and ROC-AUC = 0.8488 .
- Divergence is governed by **cis-regulatory architecture**, not coding identity.
- Four out-of-fold scores — `score_shared`, `score_rank`, `score_direction`, `score_translational` — prioritise candidate genes.

What to do next

- Match timepoints by **biological equivalence**, not chronology.
- Test **prime-boost and adjuvanted** regimens for low-immunogenicity platforms.
- Align **strain with the hypothesis** and standardise clinical metadata.
- Extend the scores beyond **blood-based BTMs** to other tissues and gene sets.
- Keep the whole framework **open and retrainable** with new multi-omic layers.

Thank you — questions welcome.